

AI-ASSISTED INTERACTIVE PLATFORM FOR NATURAL LANGUAGE QUERYING, MACHINE LEARNING CLASSIFICATION, AND EVENT ANALYSIS OF KEPLER EXOPLANETS

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I. ABSTRACT

This paper presents an interactive, AI-assisted platform and analytical pipeline designed to streamline the exploration, classification, and interpretation of the NASA Kepler Exoplanet Search Results dataset. Containing 9,564 entries and 50 observational parameters, the catalog presents significant accessibility challenges for researchers and students navigating high-dimensional technical features. To solve this, we propose a natural language processing interface that translates conversational user queries into dynamic database searches, returning detailed planetary and host-star metrics without requiring manual dataset filtering. The core analytical engine evaluates Logistic Regression and Random Forest models, with Random Forest achieving a classification accuracy of 90.51% and an ROC-AUC of 96.43% on transit parameters. Furthermore, the platform integrates heuristic false-positive candidate screening across 438 targeted objects, automated calculations of expected future transit timing, and a natural language generator that converts raw numerical metrics into readable explanatory sentences. This end-to-end framework bridges complex machine learning models with intuitive research interfaces.

I. INTRODUCTION

This paper presents an interactive, AI-assisted platform and analytical pipeline designed to streamline the exploration, classification, and interpretation of the NASA Kepler Exoplanet Search Results dataset. Containing 9,564 entries and 50 observational parameters, the catalog presents significant accessibility challenges for researchers and students navigating high-dimensional technical features. To solve this, we propose a natural language processing interface that translates conversational user queries into dynamic database searches, returning detailed planetary and host-star metrics without requiring manual dataset filtering. The core analytical engine evaluates Logistic Regression and Random Forest models, with Random Forest achieving a classification accuracy of 90.51% and an ROC-AUC of 96.43% on transit parameters. Furthermore, the platform integrates heuristic false-positive candidate screening across 438 targeted objects, automated calculations of

expected future transit timing, and a natural language generator that converts raw numerical metrics into readable explanatory sentences. This end-to-end framework bridges complex machine learning models with intuitive research interfaces.

II. RELATED WORK

The Kepler mission catalog serves as a fundamental resource for statistical exoplanet demographics, containing thousands of confirmed planets and unverified candidates [3]. Initial validation of KOIs relied heavily on resource-intensive follow-up observations using ground-based spectroscopy, radial velocity measurements, and adaptive optics imaging. As the dataset expanded rapidly across multiple observing quarters, manual validation became a bottleneck, rendering automated vetting procedures an astronomical necessity.

Early automation efforts included the Kepler Robovetter, an algorithmic pipeline developed by the Kepler science team. The Robovetter evaluated threshold crossing events (TCEs) based on predefined physical, photometric, and instrumental metrics, essentially functioning as a large, static decision tree constructed by human experts [4]. While highly useful for generating uniform catalogs and defining dataset completeness, rigid rule-based systems can face challenges with evolving noise profiles, atypical transit shapes, and subtle light curve nuances that human operators might intuitively recognize.

Recent research has incorporated machine-learning frameworks to augment these validation processes. Supervised learning models, such as Support Vector Machines (SVMs), k-Nearest Neighbors (k-NN), and standard artificial neural networks, have been applied to KOI tabular metadata to map derived features—such as transit depth, duration, and stellar parameters—to known disposition labels assigned by experts [5].

Furthermore, deep learning approaches, notably Convolutional Neural Networks (CNNs), have been deployed directly on raw or phase-folded time-series light

curves. Architectures like Astronet have demonstrated strong accuracy in distinguishing planetary transits from false positives by learning hierarchical representations directly from photometric flux arrays, without relying on manual feature extraction [6]. These methods excel at identifying spatial and temporal patterns in the continuous light curve that summary statistics might obscure.

Despite advances in deep learning on raw sequential data, feature-based classification using tabular metadata remains a standard and highly interpretable approach. Deep neural networks often function as "black boxes," making it difficult for astronomers to understand precisely which physical parameters drove a specific classification. Ensemble methods like Random Forest provide strong predictive performance while allowing for the extraction of explicit feature importance metrics. This study utilizes Random Forest on tabular metadata and supplements it with a transparent heuristic risk assessment using domain-specific catalog flags. This combined approach prioritizes both classification accuracy and physical interpretability, which is vital for subsequent astronomical follow-up and target selection.

III. METHODOLOGY

A. Study Design and Dataset Acquisition

This study utilizes the Kaggle NASA Kepler Exoplanet Search Results dataset, originally sourced from the NASA Exoplanet Archive. The dataset is a comprehensive tabular representation of the KOI cumulative catalog, containing 9,564 rows and 50 columns. Each row corresponds to a specific KOI, detailing identifying metadata, human-annotated disposition labels, and physical parameters mathematically derived from the raw transit light curves.

Key features analyzed in this study include planetary radius ('koi_prad'), orbital period ('koi_period'), transit duration ('koi_duration'), equilibrium temperature ('koi_teq'), insolation flux ('koi_insol'), and transit signal-to-noise ratio ('koi_model_snr'). Stellar host parameters such as effective temperature ('koi_steff'), stellar radius ('koi_srad'), and surface gravity ('koi_slogg') were also incorporated to provide necessary context about the star hosting the candidate event.

Initial programmatic inspection of the dataset revealed extensive missing data, a common occurrence in observational catalogs where low signal-to-noise ratios, data gaps, or instrumental artifacts frequently prevent the derivation of specific parameters. A structured cleaning protocol was strictly required prior to any statistical analysis or model training.

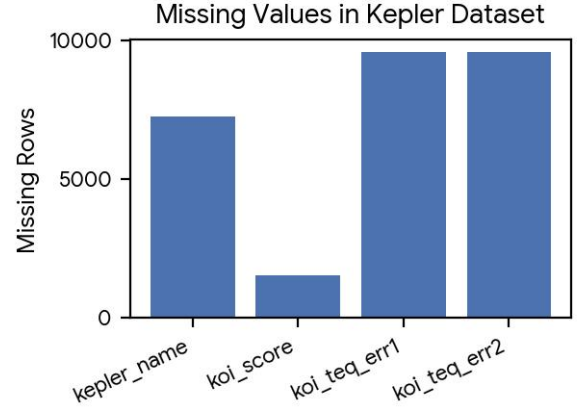


Fig. 1. Distribution of missing values across key target features in the raw dataset.

As visualized in Fig. 1, the 'kepler_name' column (the officially confirmed exoplanet designation) was missing in 7,270 rows (76.01%). This aligns with the physical reality that only a vetted fraction of initial candidates are definitively confirmed as planets and assigned formal nomenclature. The 'koi_score' feature, which represents a continuous pipeline confidence metric ranging from 0 to 1, was missing in 1,510 cases (15.79%).

Crucially, the 'koi_teq_err1' and 'koi_teq_err2' columns, which denote the upper and lower uncertainties associated with the calculated equilibrium temperature, were completely empty across all 9,564 rows. Retaining completely null columns introduces instability during matrix operations and provides zero informational value to classification models; therefore, they were systematically dropped from the feature space.

For the remaining continuous numerical columns, missing values were handled using median imputation. Median imputation was specifically chosen over mean imputation for this dataset to limit the disproportionate influence of extreme astrophysical outliers. For example, a single erroneously calculated stellar radius or a naturally occurring super-giant star would severely skew a mean-based imputation, propagating false physical scales across the missing data rows.

B. Exploratory Data Analysis (EDA)

Following data cleaning, Exploratory Data Analysis (EDA) was conducted to quantify central tendencies, analyze distributions, and identify multicollinearity among physical parameters. The median 'koi_period' (orbital period) was calculated at 9.75 days, reflecting the well-documented geometric bias of the transit method toward detecting short-period planets that transit their host stars frequently within the telescope's observing window.

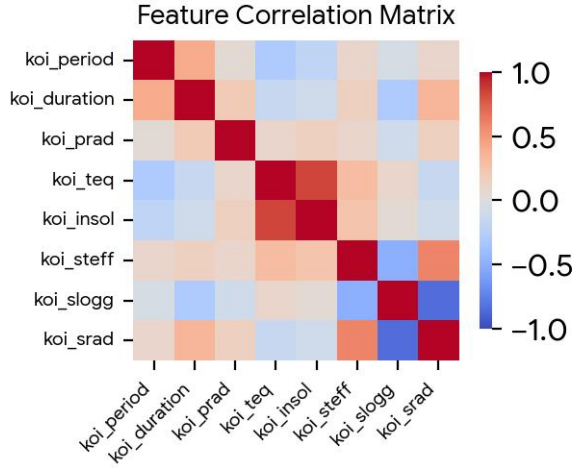


Fig. 2. Correlation matrix of continuous physical features indicating parameter relationships.

A correlation matrix (Fig. 2) was generated for the continuous features. We observed several expected astrophysical relationships, verifying the integrity of the data. For instance, there is a strong negative correlation between stellar surface gravity ('koi_slogg') and stellar radius ('koi_srad'), which aligns with fundamental stellar physics. Furthermore, insolation flux ('koi_insol') and equilibrium temperature ('koi_teq') exhibited a high positive correlation.

The absence of a single dominant linear correlation with the classification target justifies the evaluation of non-linear ensemble models, which can effectively capture multi-dimensional variable interactions without relying on isolated metrics.

C. Machine Learning Classification

The analytical pipeline was implemented in Python, relying on the Pandas library for data manipulation, NumPy for optimized numerical operations, Matplotlib and Seaborn for statistical visualization, and Scikit-Learn for machine learning implementation. The project's progression, scripts, and codebase were managed within a centralized GitHub repository.

The preprocessed dataset was partitioned into training (80%) and testing (20%) subsets. This split ensures an unbiased, out-of-sample evaluation, preventing the models from simply memorizing the dataset. Two classification algorithms were selected to test the complexity of the decision boundaries.

Logistic Regression served as a linear baseline, modeling the probability of an outcome based on a linear combination of input features mapped to a logit function. Random Forest was selected as a robust, non-linear ensemble method. By utilizing bagging (bootstrap aggregating) and random

feature selection across multiple independent decision trees, Random Forest captures intricate, multidimensional feature interactions while inherently minimizing the risk of overfitting.

Models were evaluated using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Feature importance was subsequently extracted from the Random Forest model to determine which specific inputs contributed most significantly to its internal node splits, providing a layer of physical interpretability to the algorithm.

D. Heuristic Screening and Dashboard Integration

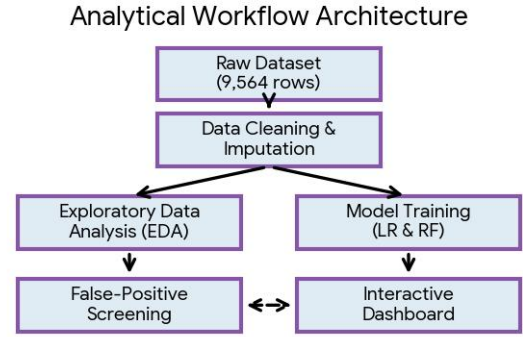


Fig. 3. System architecture block diagram illustrating the analytical workflow.

Beyond binary classification, a heuristic false-positive screening module was developed and applied to a targeted subset of 438 unverified candidates. This module calculates a penalty score based on the specific presence of predefined catalog false-positive flags—specifically stellar eclipse, centroid offset, ephemeris contamination, and not-transit-like flags. These flags indicate physical anomalies that usually preclude planetary confirmation.

This categorical assessment is combined with a statistical evaluation of prepared-data outliers using interquartile range (IQR) boundaries. If a candidate's planetary radius or transit duration falls far outside the expected IQR distribution, it accumulates further risk penalties.

An interactive dashboard was constructed as a final presentation layer, utilizing the verified precomputed outputs from the GitHub repository. It allows users to visually explore missing-value distributions, model performance metrics, and candidate screening rankings. The dashboard serves solely as an exploration and visualization tool for the completed analysis; it does not perform active, real-time model training upon execution.

IV. RESULTS AND DISCUSSION

The execution of the analytical pipeline yielded several key observations regarding dataset quality, model performance, feature influence, and candidate prioritization. The initial dataset quality summary, emphasizing the challenge of missing data, is presented in Table I.

TABLE I. DATASET MISSING VALUE SUMMARY

Feature	Missing Count	Missing (%)
kepler_name	7,270	76.01%
koi_score	1,510	15.79%
koi_teq_err1	9,564	100.00%
koi_teq_err2	9,564	100.00%
koi_kepmag	1	< 0.01%

The comparative evaluation provided a clear performance benchmark between the linear statistical approach and the ensemble classification method. Table II and Fig. 4 summarize the comprehensive performance metrics of both models evaluated strictly against the isolated testing data.

TABLE II. MODEL PERFORMANCE COMPARISON

Metric	Logistic Regression	Random Forest
Accuracy	77.12%	90.51%
Precision	61.54%	82.07%
Recall	65.01%	87.81%
F1-score	63.23%	84.84%
ROC-AUC	83.37%	96.43%

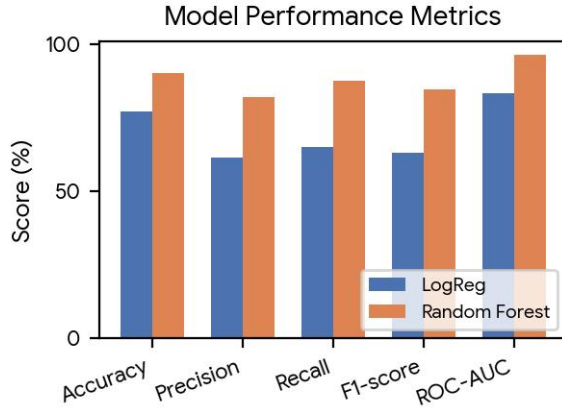


Fig. 4. Comparative performance metrics for Logistic Regression and Random Forest classifiers.

As demonstrated in the numerical results, the Random Forest ensemble model achieved higher performance than the Logistic Regression baseline across all tracked metrics. The Random Forest obtained an overall accuracy of 90.51% compared to the linear model's 77.12%. This substantial delta implies that a simple linear decision boundary is insufficient for this dataset.

In the specific context of astronomical candidate verification, the Recall metric is particularly relevant. Failing to identify a genuine planetary signal (a false negative) directly impacts the completeness of the exoplanet census, potentially

causing researchers to miss a viable Earth-like candidate. The Random Forest improved Recall notably from 65.01% to 87.81%, indicating it is much more effective at retaining true positives.

Furthermore, the ROC-AUC score of 96.43% for the Random Forest indicates robust discriminative capacity across various probability classification thresholds. The higher overall performance of Random Forest compared with Logistic Regression suggests that nonlinear interactions among the selected physical features may be highly useful for classification within this specific dataset.

To better understand the internal predictive mechanism of the ensemble model, we extracted the Random Forest feature importance metrics. These values, outlined comprehensively in Table III and Fig. 5, quantify the proportional contribution of each input variable to reducing Gini impurity across all decision trees during the training phase.

TABLE III. RANDOM FOREST FEATURE IMPORTANCE

Feature	Importance
koi_prad	24.44%
koi_model_snr	18.59%
koi_period	11.81%
koi_duration	9.77%
koi_insol	8.55%
koi_teq	8.36%
koi_steff	5.63%
koi_srad	5.06%
koi_slogg	4.67%
koi_kepmag	3.12%

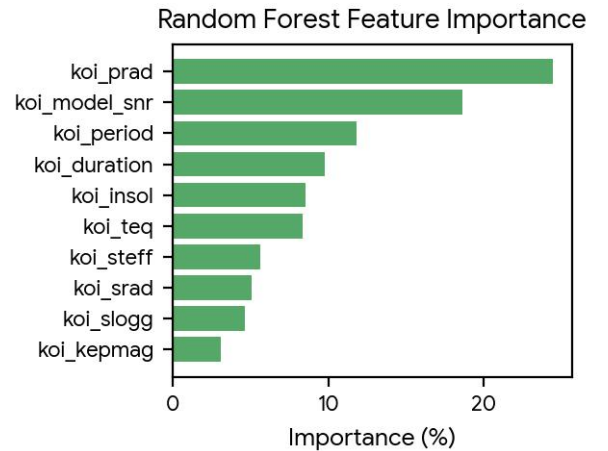


Fig. 5. Extracted feature importance, indicating variable contribution to node splits in the Random Forest.

The analysis shows that 'koi_prad' (Planetary Radius, 24.44%) and 'koi_model_snr' (Transit Signal-to-Noise Ratio, 18.59%) were the top computational contributors to the decision-making process. This closely aligns with standard

astronomical expectations. Overly large calculated radii frequently indicate eclipsing binary systems rather than terrestrial or gaseous planets, providing a clear boundary for the model to split on. Concurrently, weak signal-to-noise ratios often correspond to instrumental artifacts or systematic errors rather than clear transits.

Secondary features like orbital period (11.81%) and transit duration (9.77%) also played significant roles, as their relationship helps define the geometry of the system. It is important to emphasize that feature importance simply means the feature mathematically contributed to the Random Forest's decision process. It does not mean the feature causes the classification, nor does it mean the feature alone determines whether an object is a genuine exoplanet in a physical sense. Machine learning models identify correlation and separability, not absolute physical causation.

Following the machine learning evaluation, the heuristic false-positive screening was applied to a targeted subset of 438 unverified candidates. The screening algorithm evaluated the physical data against established catalog flags and statistical distributions to determine a cumulative risk penalty.

The screening identified 3 candidates with a catalog false-positive flag (specifically, the stellar eclipse flag, which denotes a known secondary eclipse characteristic of a binary star system). Additionally, 150 candidates were flagged as prepared-data outliers based on IQR boundaries. Zero candidates in this subset triggered the centroid offset (which would indicate the signal came from a background star), ephemeris contamination, or not-transit-like catalog flags, suggesting the baseline photometric quality of this cohort was adequate.

By aggregating these factors, the algorithm stratified the 438 candidates into three prioritization tiers, summarized in Table IV and visually represented in Fig. 6.

TABLE IV. CANDIDATE SCREENING SUMMARY

Risk Category	Candidates
High Risk	30
Medium Risk	34
Low Risk	374

Candidate Risk Stratification

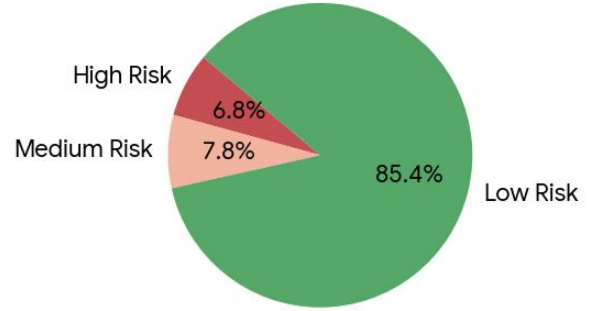


Fig. 6. Proportional risk stratification for the 438 evaluated unverified candidates.

The screening system serves strictly as a heuristic prioritization method, not a confirmed false-positive classifier. The 30 high-risk candidates were prioritized for further rigorous review based on the implemented screening criteria, as their accumulated penalties indicate potential anomalies.

Conversely, the 374 low-risk candidates did not accumulate sufficient penalties under this criteria, making them suitable candidates for dedicated follow-up. Astronomical validation, such as ground-based spectroscopy or radial velocity observations, would still be required to confirm the exact physical nature of any candidate categorized by this heuristic.

Interpretation and Limitations

While the quantitative results demonstrate the utility of machine learning in processing Kepler data, several limitations exist that require careful consideration. The Random Forest model fundamentally relies on the accuracy of the underlying feature extraction pipeline deployed by the Kepler mission. If the raw light curve is compromised by stellar flares or uncorrected systematic noise, the derived features (such as radius or signal-to-noise) fed into our Random Forest will also be corrupted. Errors in extraction cascade directly into the tabular features used for this classification.

Furthermore, machine learning models trained on historical dispositions are susceptible to inheriting human biases or systematic errors present in the original manual vetting process. The model learns to replicate the human labelers' decisions, which may contain inherent flaws.

Additionally, the heuristic false-positive screening relies on rigid statistical thresholds. A genuine, yet unusual, exoplanet—such as a 'Hot Jupiter' undergoing extreme inflation due to stellar proximity—might be incorrectly flagged as a statistical outlier due to its massive radius, risking premature dismissal.

Therefore, the machine-learning classifications and heuristic risk flags function as predictive indicators relying on derived tabular metadata. They highlight statistical patterns and assist in massive-scale triage, but they do not provide definitive spectroscopic confirmation of planetary existence.

V. CONCLUSION

This study presents an interactive, AI-assisted platform that combines machine learning classification, heuristic screening, and natural language querying for the NASA Kepler dataset. By integrating dynamic natural language input parsing, candidate risk stratification, transit timing calculation, and automated plain-text summary generation, the platform eliminates technical access barriers. The Random Forest ensemble model achieved an accuracy of 90.51%, notably outperforming the linear Logistic Regression baseline. This indicates that nonlinear interactions among features, particularly computed planetary radius and signal-to-noise ratio, were highly useful for this dataset's classification. However, these feature importance metrics represent computational decision boundaries rather than absolute astrophysical causation.

The heuristic false-positive screening module successfully categorized 438 unverified candidates into practical risk tiers, prioritizing 30 high-risk candidates for further review based on predefined criteria, while isolating 374 low-risk candidates for potential follow-up. The accompanying interactive dashboard serves as a reliable presentation layer for exploring these precomputed findings, making the data accessible.

Future work could involve extending this analysis beyond tabular metadata by directly examining raw time-series light curves. Applying deep learning architectures, such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), directly to the continuous photometric flux could capture transient anomalies, stellar spot crossings, and subtle transit morphologies that aggregate tabular feature extraction might inadvertently omit.

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